COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE 21S3
DATE PERFORMED:	10/20/2026	**DATE SUBMITTED:**	10/20/2026
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ACTIVITY NO. 14.1 Algorithm Complexity

1. PROCEDURE OUTPUT

	Best Case	Worst Case	Analysis
Algorithm 1 (Bubble Sort)	O(n)	O(n2)	Bubble Sort requires nested loops resulting in quadratic growth while Merge sort strictly adheres to logarithmic division and linear merging
Algorithm 2 (Merge Sort)	O(n log n)	O(n log n)	Merge Sort consistently splits and merges regardless of the initial order, while bubble sort can break early for the best case scenario.

Table 14-1. Best- and Worst-Case Analysis using Theoretical Tools

	Algorithm 1(Bubble Sort)	Algorithm 2 (Merge Sort)
Best Case	**n = 100: 0.0315ms** **n = 1000: 2.4648ms** **n = 10000: 269.6190ms**	**n = 100: 0.0612 ms** **n = 1000: 0.6069ms** **n = 10000: 6.7847 ms**
Worst Case	**n = 100: 0.0319 ms** **n = 1000: 3.5598 ms** **n = 10000: 297.3777 ms**	**n = 100: 0.0641 ms** **n = 1000: 0.5764 ms** **n = 10000: 6.7568 ms**

Table 14-2. Best- and Worst-Case Analysis using Experimental Tools

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO A: Contrast and compare the two searching techniques used in class. The required deliverable for this activity would be the theoretical and empirical analysis.
For the theoretical analysis, the deliverable must be:

- Pseudocode Analysis
- Time and Space Complexities

For the empirical analysis, the deliverable must be the filled-out table and your analysis.

Input Size (N)	Elapsed Time for Binary Search (Worst-Case)	Elapsed Time for Linear Search (Worst-Case)
1,000	0.0002	0.0026
10,000	0.0002	0.0147
100,000	0.0002	0.1529

Provide your analysis:
Based on the empirical results, Linear Search's execution time grew linearly from 0.0026ms to 0.1529ms as input size N increased to 100,000, whereas Binary Search maintained a virtually flat logarithmic runtime of 0.0002ms across all test sizes because it cuts the search space in half at each step.

Conclusion:
Busa: This lab activity proved how theoretical algorithm complexity directly impacts real-world program runtime. During our sorting tests, Bubble Sort became extremely slow as data increased, reaching 297.3777ms at n=10,000 due to its quadratic O(n^2) complexity. The Merge Sort stayed fast at around 6.7568ms because it uses a divide and conquer approach with O(n log n)time complexity. For searching, Binary Search clearly outperformed Linear Search on larger sorted arrays. Choosing the right algorithm structure is very important for writing efficient codes .

Geronimo: Our results prove that choosing a good algorithm matters way more than hardware speed. Bubble Sort scaled poorly up to 297.3777ms at N=10,000 while Merge Sort finished in just 6.7ms using O(nn). The Binary Search saved a lot of time compared to Linear Search on large datasets. Understanding Big-O complexity helps us write better and more scalable programs.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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